Configure IP-Based Routing Policy in Amazon Route 53

Summary:-

In this lab, an IP-Based Routing Policy is configured in Amazon Route 53. CIDR Collections are created to associate specific source IP address ranges with different DNS responses. DNS records are then configured to return different IP addresses based on the source IP address of the DNS resolver. The routing policy is verified using both **nslookup** and the **Test Record** feature.

Let's Start:-

- Create a CIDR Collection in Amazon Route 53.
- Add CIDR locations for different source IP ranges.
- Configure IP-Based Routing Policy records.
- Associate each record with a CIDR location.
- Verify the routing policy using DNS queries.

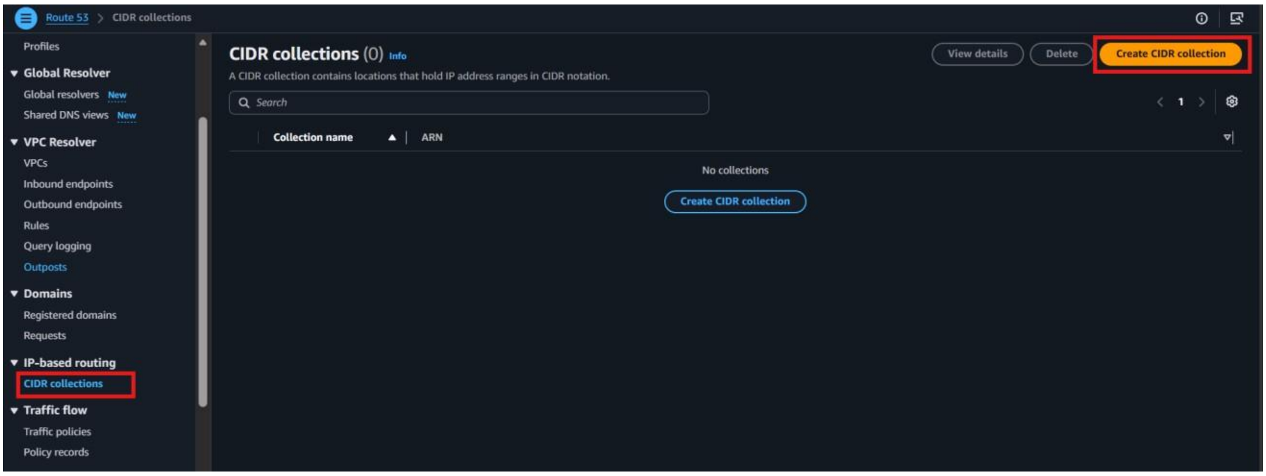
Lab Steps:-

Step 1 – Verify the Environment

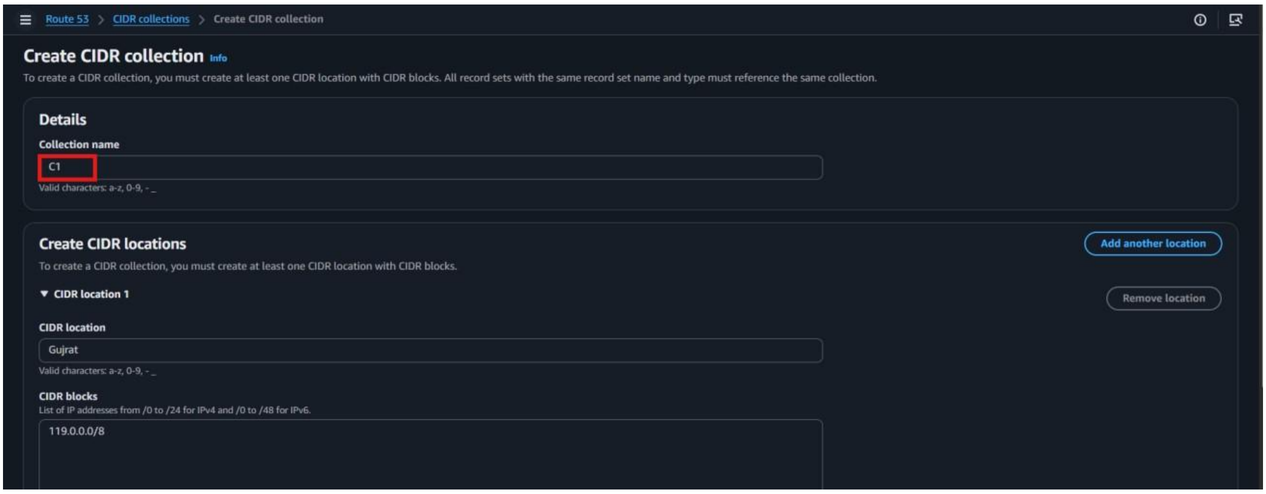
- Open the **Amazon Route 53** console.
- Select an existing **Public Hosted Zone**.
- This step prepares the required Route 53 resources for the IP-Based Routing Policy.

Step 2 – Create a CIDR Collection

- Open the **Amazon Route 53** console.
- Select **CIDR Collections**.
- Click **Create CIDR Collection**.



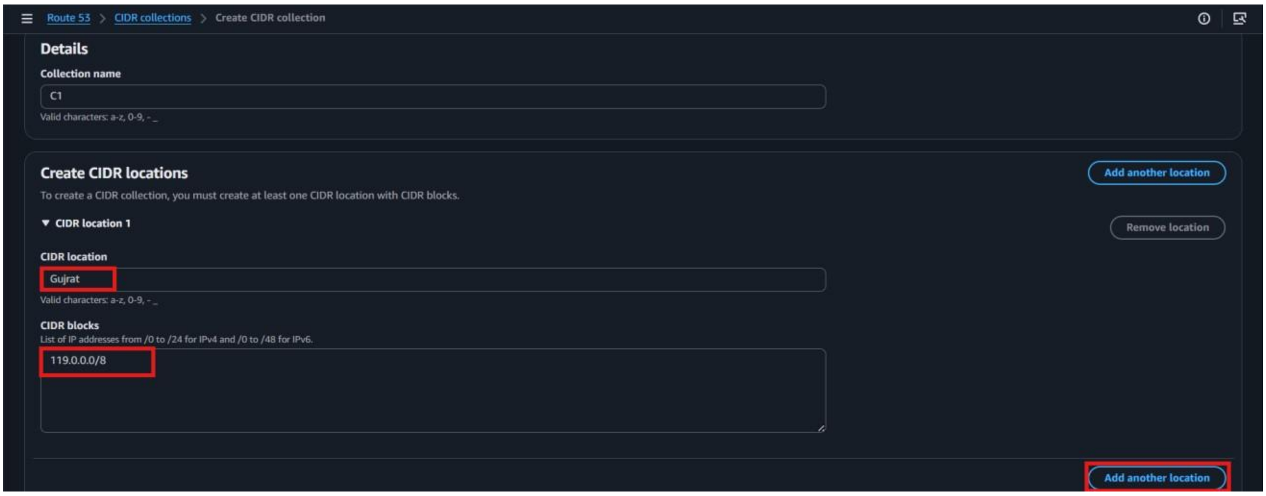
- Enter the following:
    - **Collection Name:** C1



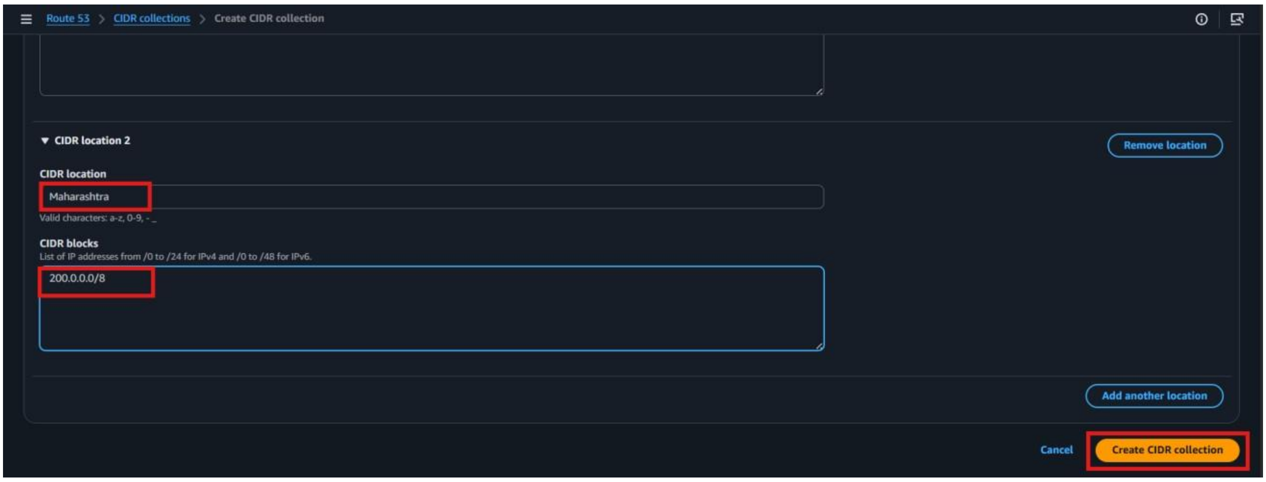
- This step creates a collection to store CIDR locations.

Step 3 – Add CIDR Locations

- Add the following CIDR locations:
    - **Location:** Gujarat
    - **CIDR:** 119.0.0.0/8



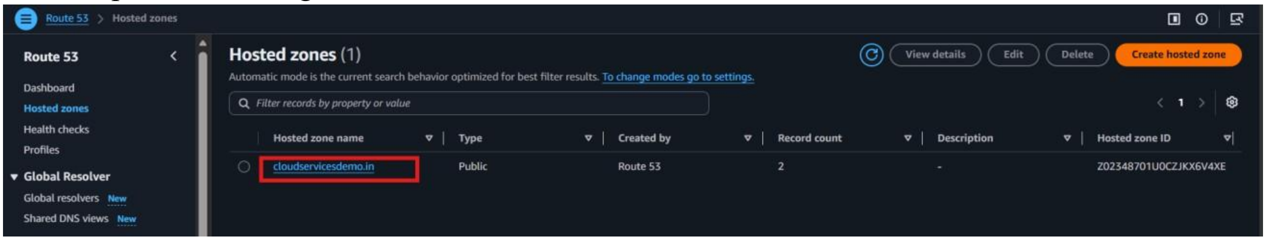
- **Location: Maharashtra**
- **CIDR: 200**.0.0.0/8



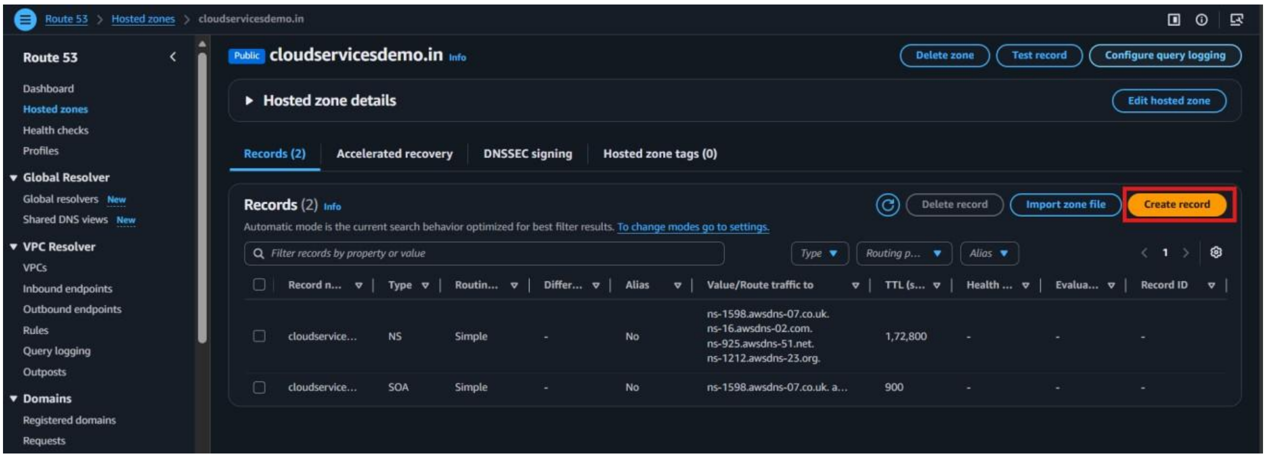
- Click **Create CIDR Collection**.
- This step associates source IP ranges with different CIDR locations.

Step 4 – Create the Gujarat Record

- Open the existing **Public Hosted Zone**.

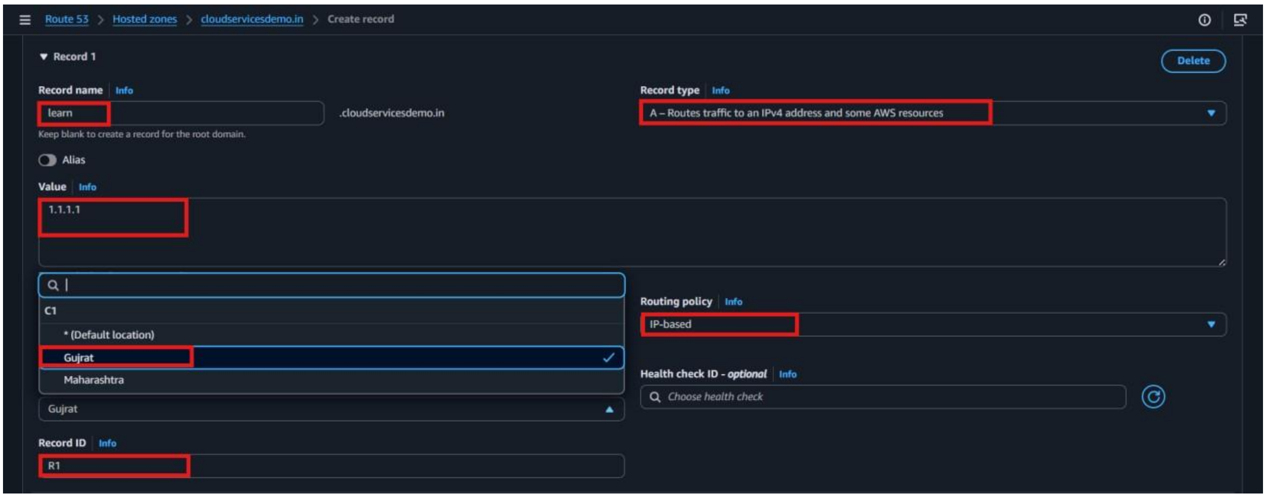


- Click **Create Record**.



- Configure the following:
  - **Record Name:** learn
  - **Record Type:** A
  - **Value:** 1.1.1.1
  - **Routing Policy:** IP-Based
  - **CIDR Collection:** C1

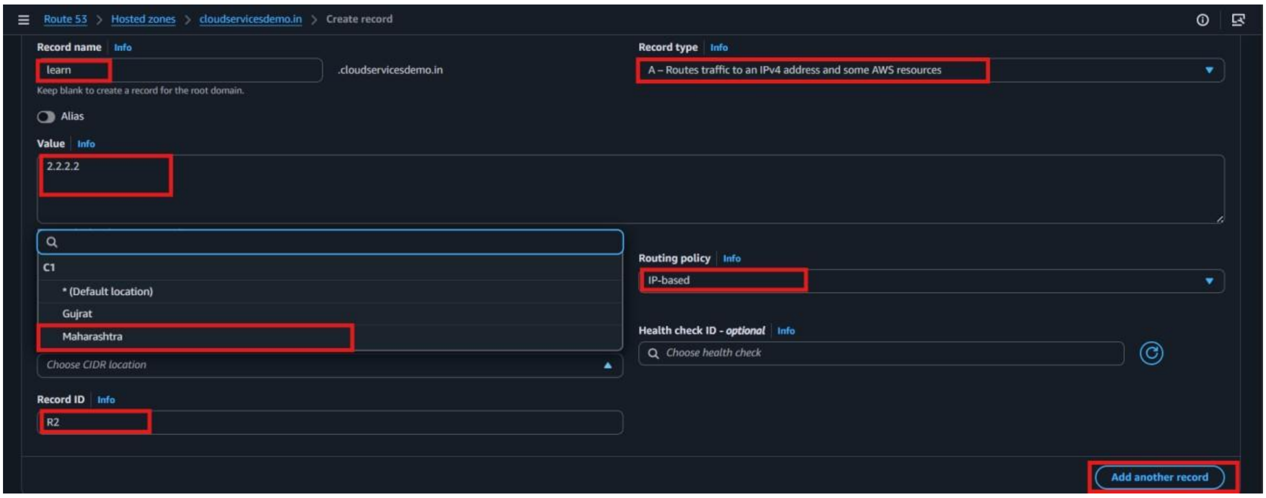
- ○ **Location:** Gujarat
- ○ **Record ID:** R1



- Click **Create Records**.
- This step returns the Gujarat endpoint for the configured source IP range.

Step 5 – Create the Maharashtra Record

- Click **Create Record**.
- Configure the following:
  - ○ **Record Name:** learn
  - ○ **Record Type:** A
  - ○ **Value:** 2.2.2.2
  - ○ **Routing Policy:** IP-Based
  - ○ **CIDR Collection:** C1
  - ○ **Location:** Maharashtra
  - ○ **Record ID:** R2

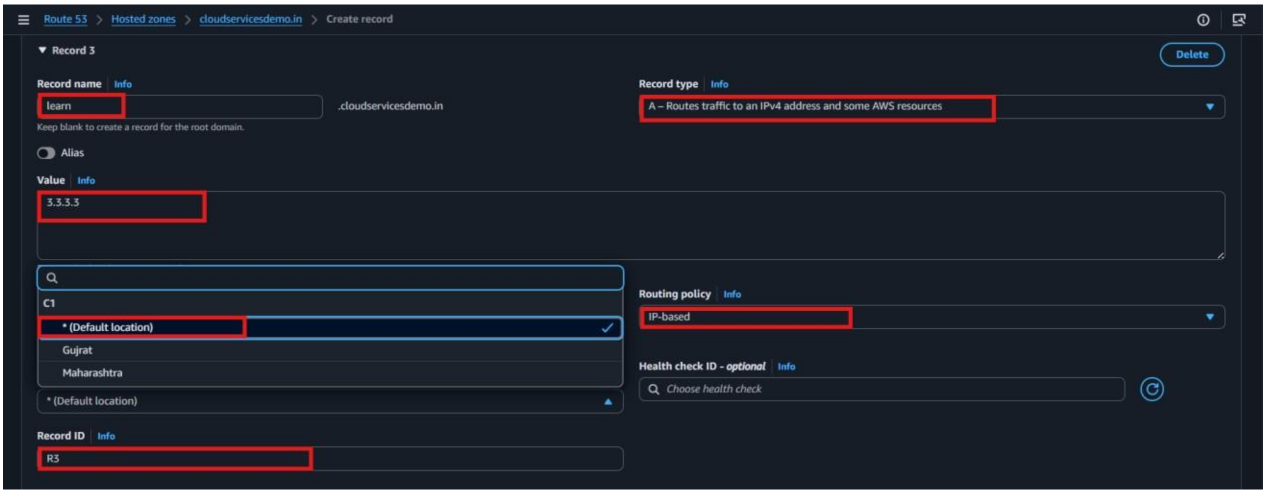


- Click **Create Records**.

- This step returns the Maharashtra endpoint for the configured source IP range.

Step 6 – Create the Default Record

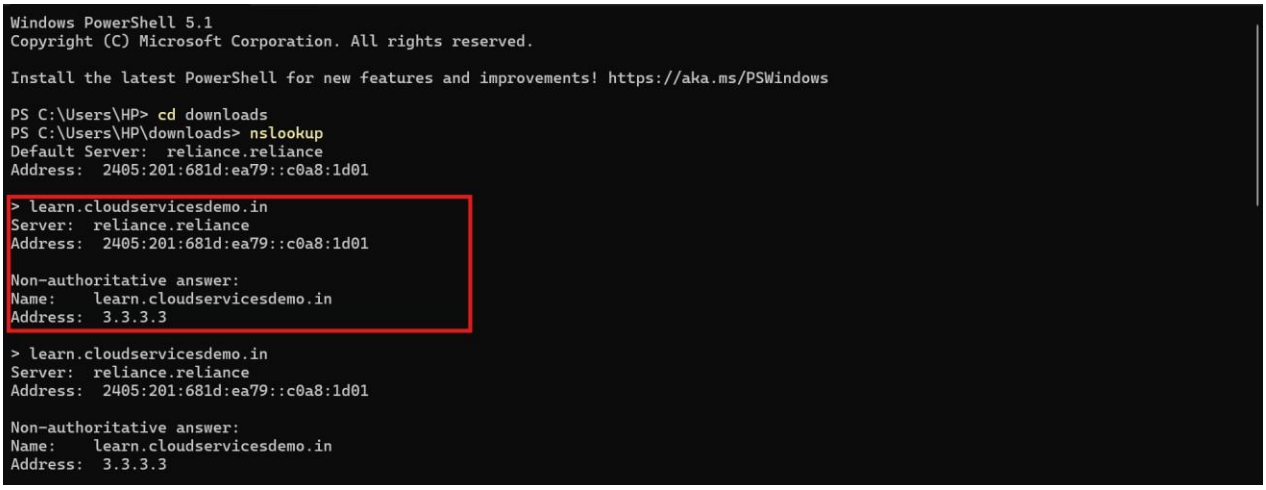
- Click **Create Record**.
- Configure the following:
    - **Record Name:** learn
    - **Record Type:** A
    - **Value:** 3.3.3.3
    - **Routing Policy:** IP-Based
    - **Location:** Default
    - **Record ID:** R3



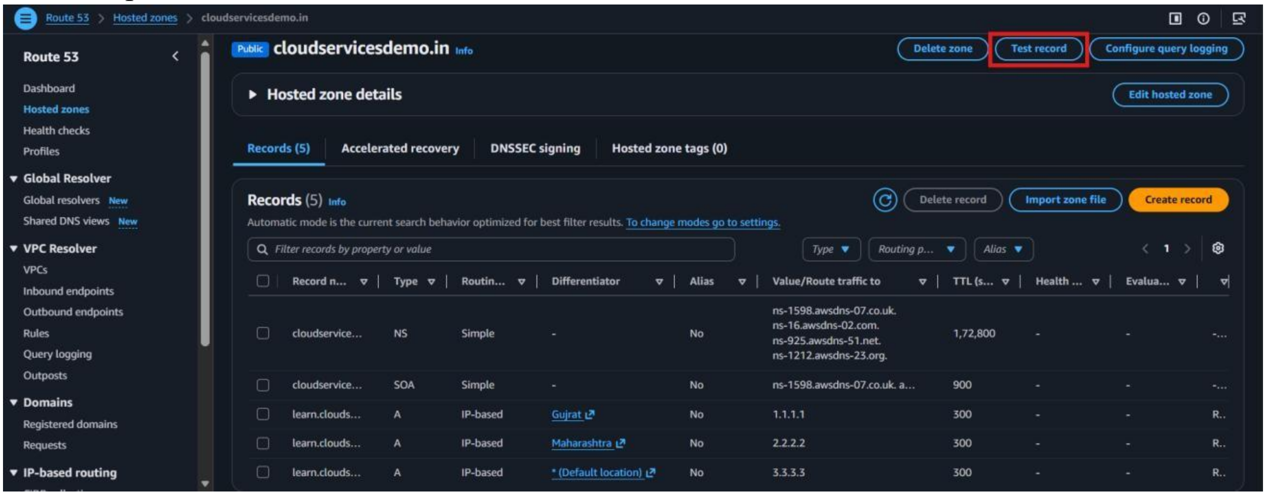
- Click **Create Records**.
- This step provides a default DNS response for all other source IP addresses.

Step 7 – Verify the IP-Based Routing Policy

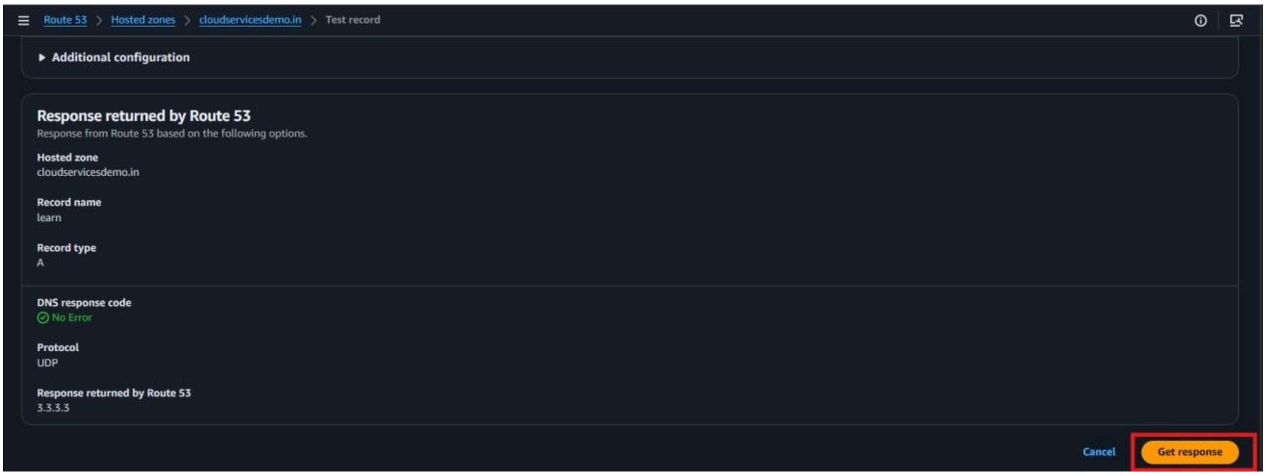
- Open **Command Prompt**.
- **Run**:
    - nslookup learn.<**your-domain-name**>



- Verify that the returned IP address matches the configured source IP range.
- Alternatively, use **Test Record** in the Hosted Zone to verify the default routing response.



- This step verifies that Route 53 returns different IP addresses based on the source IP address.



Verification

- Successfully created a CIDR Collection.
- Added multiple CIDR locations.
- Configured IP-Based Routing Policy records.
- Associated DNS records with CIDR locations.
- Verified source IP-based DNS resolution using **nslookup** and **Test Record**.